

GATE 0 - PROJECT SYNOPSIS

Themes 1-5 | System / Tool / Engine based Mini Projects

Expected length: about 4 pages + cover page

Team No.	1
Theme No.	1- Data storage and retrieval system
Project Title	Mini RDBMS Engine with Query Processing & Transaction Support
With Guide Name	Ms. Jyoti Patil
Team Members	1. Bhoomi B 2. Sneha S 3. Soumya Patil 4. Subham M

Student Preparation Instruction

This synopsis is intended to demonstrate the team's own understanding of the proposed project. Students should study the project, discuss within the team, consult relevant textbooks/technical resources, and prepare the content in their own words. AI tools shall not be used to generate the contents of this document.

At Gate 0, do NOT finalize: detailed functional/non-functional requirements, architecture/modules, algorithms/data structures, APIs, technology stack, alternate designs, detailed KPIs or test cases.

1. Background & Motivation

Briefly describe the proposed system/tool/engine, its purpose, broad functionalities/computational capabilities, and why developing it is technically meaningful as a Mini Project. Highlight opportunities to apply and integrate foundational Computer Science concepts.

RDBMS Systems are foundational to all modern software yet students interact with it the only as end users. Core engine working of the existing engines is unknown This project proposes building a simplified RDBMS engine capable of storing structured data, processing SQL-like queries, and supporting transactions with basic correctness guarantees. This project has a integrated approach as it include concepts from DAA, OSPP, DBMS AND POCD (majorly). It gives the team hands-on exposure to how real-world systems balance correctness, efficiency, and reliability, bridging the gap between theoretical coursework and practical systems engineering.

2. Project Objectives

State 3-5 broad engineering objectives. Describe what the team intends to develop or study. Do not prematurely prescribe architecture, algorithms, data structures, APIs or implementation mechanisms.

1. To design and implement a system for structured storage and retrieval of relational data.
2. To develop a mechanism for interpreting and executing user-issued queries on the stored data.
3. To incorporate transaction handling that preserves data correctness during sequences of operations.
4. To study and address the challenges of maintaining consistency when multiple operations occur in sequence or concurrently.
5. To evaluate the system's correctness and performance through a set of representative test scenarios.

FOUNDATION COURSE INTEGRATION & CONCEPTUAL VIEW

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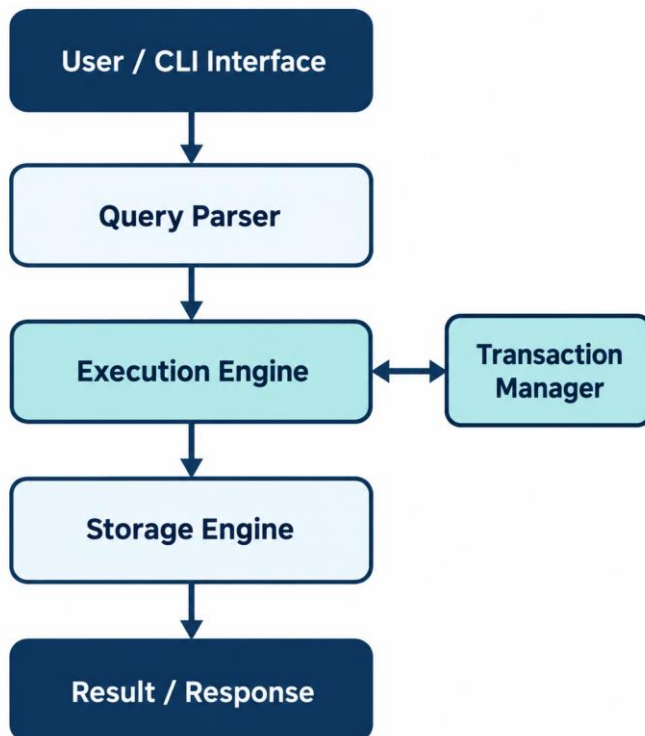
3. Foundation Course Integration

Identify at least three foundational courses that are expected to contribute meaningful technical depth. Mention only broad concepts/areas at this stage.

Sl. No.	Foundation Course	Broad Concepts / Areas Expected to be Applied	Role
1	Database Management Systems	Relational model, SQL query semantics, indexing, ACID/transaction theory	Dominant
2	Data Structures & Algorithms	Efficient storage structures (hash maps, trees) for tables; algorithmic complexity of scan/search/filter operations	Dominant
3	Operating Systems Principles	File I/O for persistence, concurrency control, locking for simultaneous transactions	Supporting
4	Object-Oriented Programming	Modular class design for parser, execution engine, storage engine, and transaction manager as separate objects	Supporting

4. Conceptual Workflow / System View

Draw one high-level block/workflow diagram showing the broad functional flow or major interactions in the proposed artefact. Keep it conceptual; detailed architecture, finalized modules, algorithms, data structures, APIs and technologies belong to later stages.



Insert / draw conceptual workflow or system view here

SCOPE & FEASIBILITY

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5. Broad Project Scope

Define the tentative technical boundary of the project. Detailed requirements will be derived later in S1.

5.1 Broadly In Scope

#	Broad In-Scope Item
1	Implement a limited SQL-like query language with CREATE TABLE, INSERT, SELECT, UPDATE, DELETE, and WHERE, using local file-based relational storage.
Core RDBMS Engine	
2	Implement lexer, parser, AST, semantic validation, and meaningful syntax/semantic error messages with basic column-name suggestions.
Query Processing & Error Explanation	
3	Provide preview/confirmation for UPDATE and DELETE, showing affected rows and proposed changes before modification.
Safe Data Modification:	
4 Transaction Support & Execution Tracing	Implement BEGIN, COMMIT, ROLLBACK, basic undo logging/recovery, and lightweight query execution tracing.

5.2 Out of Scope

#	Out-of-Scope Item	Reason
1	Full SQL Standard & Advanced Query Features	Supporting the complete SQL language, advanced joins, subqueries, window functions, triggers and stored procedures is beyond a 3-credit mini-project.
2	Distributed Database, Replication & Cloud Deployment	Requires networking, distributed consistency, failure handling and multiple systems; unnecessary for the core objective.
3	Advanced Concurrency & Optimization	MVCC, multiple isolation levels, advanced cost-based optimization, hybrid row-column storage and large-scale performance tuning require substantially more implementation and testing.

6. Feasibility & Constraints

Briefly assess whether the proposed scope is achievable as a 3-credit Mini Project. Identify required resources, prerequisite knowledge and major constraints. Existing libraries/tools may support peripheral functionality but should not completely hide the core mechanisms students are expected to design, implement and study.

Feasibility within 3-credit scope	The project uses a limited SQL grammar, single-node local storage, basic query execution, semantic checking, safe modification, transaction management and lightweight recovery. Advanced SQL, distributed processing, MVCC and production-scale optimization are excluded.
Required computing / resources	C++, VS Code, Git/GitHub, local file storage and sample datasets. No GPU, cloud server or specialized hardware

	required.
Prerequisite knowledge	C++, Data Structures & Algorithms, DBMS fundamentals, Compiler Design basics, Operating System file I/O and basic transaction concepts.
Major constraints / limitations	15-week academic timeline, student-level implementation experience, limited SQL grammar, local file-based storage, single-node execution, small-to-medium test datasets, and limited transaction/concurrency guarantees compared with production DBMSs.
Use of existing libraries / tools	Standard libraries may be used for file handling, timing and basic utilities, but the core mechanisms—query tokenization, parsing, AST construction, semantic validation, query execution and transaction logic—will be implemented by the team rather than hidden behind an external database engine.
Overall feasibility	Feasible with controlled scope. The core RDBMS, query processing, error explanation, safe modification, transaction support and execution tracing are achievable within the 3-credit project. Advanced SQL, distributed processing, MVCC and production-scale optimization remain outside the scope.

INITIAL PROJECT RISKS & FACULTY REVIEW

Gate 0 - Themes 1-5

7. Initial Project Risks

Identify 2-3 major risks that may affect completion of the project and briefly state how the team plans to address them. These are initial risks and may be revised as the project evolves.

Sl. No.	Possible Risk	How will you address it?
1	Rollback leaves inconsistent data	Early, repeated rollback testing
2	Deadlocks/race conditions in concurrency	Simple locking mechanism, test under simulated concurrent access
3	Timeline too tight for full ACID support	Build MVP first, treat extras as stretch goals
4	Modules fail to integrate cleanly	Define module interfaces before coding separately
5	Uneven team skill on modules	Pair members, document code for handoff

. Gate 0 - Faculty Review

- ☐ Technical motivation and objectives are clear.
- ☐ At least three foundation courses show potential for meaningful technical depth.
- ☐ Conceptual workflow/system view clearly communicates the proposed artefact.
- ☐ Broad scope is appropriate for a 3-credit Mini Project.
- ☐ Core mechanisms can be designed/implemented rather than fully hidden by ready-made tools.
- ☐ Feasibility, constraints and initial risks are reasonably identified.

Decision: ☐ Approved ☐ Approved with Modifications ☐ Resubmit

Faculty Remarks:

Guide Signature: _____ Date: _____